EDUKRIPA

CLASS - 10 CBSE **MATH TEST - 4** **Marks: 40**

Section - A

1. If the square of difference of the zeroes of the quadratic polynomial $x^2 + px$ + 45 is equal to 144, then the value of p is

   (a) ± 9 (b) ± 12 (c)± 15 (d) ± 18

2. ΔABC and ΔBDE are two equilateral triangle such that D is the mid-point of BC . Ratio of the areas of triangles ABC and BDE is ..... .

   (a) 1 : 4 (b) 4 : 1 (c) 1 : 3 (d) 3 : 1

3. The quadratic equation $x^2 - 4x - 3\sqrt{2} = 0$ has

   (a) two distinct real roots (b) two equal real roots

   (c) no real roots (d) more than 2 real roots

4. If $x^2 + y^2$= 25, xy = 12, then x is

   (a) (3 ,4) (b) (3, -3) (c) (3, 4, -3,- 4) (d) (3, -3)

5. The pair of equations x = a and y = b graphically represents lines which are

   (a) parallel (b) intersecting at (b, a) (c) coincident (d) intersecting at (a, b)

6. From a solid circular cylinder with height 10 cm and radius of the base 6 cm, a right circular cone of the same height and same base is removed, then the volume of remaining solid is

   (a) 280π cm^3 (b) 330π cm^3 (c) 240π cm^3 (d) 440π cm^3

7. If median is 137 and mean is 137.05, then the value of mode is

   (a) 156.90 (b) 136.90 (c) 186.90 (d) 206.90

8. What are the values of x and y for the following pair of linear equations ?

   99x y + 101 = 499 and 101 9 x y + 9 = 501

   (a) 3 and 6 (b) 3 and 2 (c) 2 and 3 (d) 6 and 3

9. The zeroes of polynomial p(x) = ax^2 + bx + c are reciprocal of each other if

   (a) b = 2a (b) c = b (c) b = a (d)c = a

10. If -1 is a zero of the polynomial p(x) = kx^2 − 4x + k, the value of k is

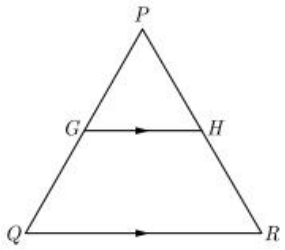
    (a) –4 (b) –2 (c) 2 (d) 4

Section - B

11. Given that HCF (306, 1314) = 18. Find LCM (306, 1314).

12. If one of the zeroes of the quadratic polynomial f(x) = $14x^2 - 42k^2x - 9$ is negative of the other, find the value of 'k '.

13. In the given figure, G is the mid-point of the side PQ of ΔPQR and GH||QR. Prove that H is the mid-point of the side PR or the triangle PQR.



Section - C

14. Solve for x : $\frac{1}{x+4} - \frac{1}{x+7} = \frac{11}{30}$, $x \neq -4, -7$.

15. From a solid right circular cylinder of height 14 cm and base radius 6 cm, a right circular cone of same height and same base removed. Find the volume of the remaining solid.

16. A metallic cylinder has radius 3 cm and height 5 cm. To reduce its weight, a conical hole is drilled in the cylinder. The conical hole has a radius of $\frac{3}{2}$ cm and its depth $\frac{8}{9}$ cm. Calculate the ratio of the volume of metal left in the cylinder to the volume of metal taken out in conical shape.

17. Write the smallest number which is divisible by both 306 and 657.

18. The mean of the following distribution is 48 and sum of all the frequency is 50. Find the missing frequencies x and y .

Class	20-30	30-40	40-50	50-60	60-70
Frequency	8	6	X	11	Y

OR

A person on tour has Rs. 4200 for his expenses. If he extends his tour for 3 days, he has to cut down his daily expenses by Rs. 70. Find the original duration of the tour.

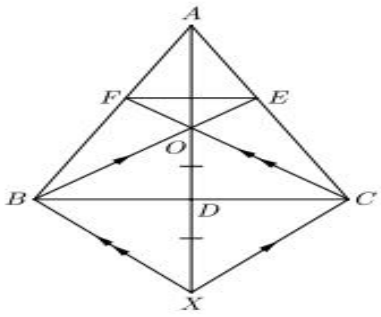
Section - D

19. In ΔABC, AD is a median and O is any point on AD. BO and CO on producing meet AC and AB at E and F respectively. Now AD is produced to X such that OD = DX as shown in figure.

Prove that :

(1) EF || BC

(2) AO : AX = AF : AB



Section - E

20. Rajesh has been given the task of designing a conical bottom tank for his client. Height of conical part is equal to its radius. Length of cylindrical part is the 3 times of its radius. Tank is closed from top. The cross section of conical tank is given below.



(i) If radius of cylindrical part is taken as 3 meter, what is the volume of above conical tank ?

(ii) What is the area of metal sheet used to make this conical tank ? Assume that tank is covered from top.

(iii) What is the ratio of volume of cylindrical part to the volume of conical part?

(iv) The cost of metal sheet is Rs. 2000 per square meter and fabrication cost is 1000 per square meter. What is the total cost of tank ?